13 THE SOURCE OF TIME

Perhaps God has many more seasons
in store for us—
or perhaps the last is to be
this winter
that guides back the waves
of the Tyrrhenian Sea
to break against
the rough pumice cliffs.
You must be wise. Pour the wine
and enclose in this brief circle
your long-cherished hope. (I, 11)

We started out with the image of time that is familiar to us: something that flows uniformly and equally throughout the universe, in the course of which all things happen. With the idea that there exists throughout the cosmos a present, a "now" that constitutes reality. The past for everyone is fixed, is gone, having already happened. The future is open, yet to be determined. Reality flows from the past, through the present, toward the future—and the evolution of things between past and future is intrinsically asymmetrical. This, we thought, is the basic structure of the world.

This familiar picture has fallen apart, has shown itself to be only an approximation of a much more complex reality.

A present that is common throughout the whole universe does not exist (chapter 3). Events are not ordered in pasts, presents, and futures; they are only "partially" ordered. There is a present that is near to us, but nothing that is "present" in a far-off galaxy. The present is a localized rather than a global phenomenon.

The difference between past and future does not exist in the elementary equations that govern events in the world (chapter 2). It issues only from the fact that, in the past, the world found itself subject to a state that, with our blurred take on things, appears particular to us.

Locally, time passes at different speeds according to where we are and at what speed we ourselves are moving. The closer we are to a mass (chapter 1),